

DEDICATED OUTSIDE AIR UNIT WITH ENERGY RECOVERY SCHEDULE

UNIT NO	SERVES	SUPPLY FANS				EXHAUST FAN			SUMMER RECOVERY				WINTER RECOVERY				DX COOLING COIL				HOT WATER COIL							ELECTRICAL DATA				MAX UNIT WEIGHT (LBS)	BASIS OF DESIGN	NOTES
		CFM MAX	FAN QTY	ESP IN WC	FAN HP EA	CFM MAX/MIN	ESP IN WC	FAN HP	SUPPLY		EXHAUST		SUPPLY		EXHAUST		ENTERING AIR DB/WB °F	LEAVING AIR DB/WB °F	TC/SC MBH	REFRIG TYPE	EAT °F	LAT °F	GPM	WPD FT WC	EWT °F	LWT °F	MBH	VOLTS/ PHASE	FLA	MCA	MAX FUSE			
									EAT DB/WB °F	LAT DB/WB °F	EAT DB/RH	LAT DB/WB	EAT °F	LAT °F	EAT@ RH °F	LAT °F																		
DOAS-1	1ST & 2ND FLRS	3770	2	1.5	4.02	3260	1.25	6.03	90°/72.0°	83.6/68.9	80.0/67.0	87.3/70.7	0.0	42.3	70.0	19.0	83.6/68.9	50.3/49.6	228.7/143.1	R454B	0.0	70.0	18.7	1.2	180	150	280.9	460/3	16.4	17.9A	20.0	3200	TRANE CSAA010	1,2,3,4,5,6,7,8,9,10,11
NOTES:		1. PROVIDE LOW LEAKAGE, MOTORIZED DAMPERS FOR OUTSIDE AIR AND EXHAUST. 2. PROVIDE THE FOLLOWING FEATURES: DOUBLE WALLED PANELS R-13 FOAM INJECTED; GALVANIZED INTERIOR PANELS; 8" INTEGRAL BASE FRAME; UL LISTED; SPRING ISOLATION; DUCT FLANGES. 3. MERV 13 OUTSIDE AIR FILTER, WITH MERV 8 PRE-FILTER, MERV 8 RETURN AIR FILTER.										4. SEE DIAGRAM C1/M-505 FOR BASIS OF DESIGN UNIT LAYOUT AND LIST OF MODULES. UNIT TO BE SHIPPED AND BROUGHT INTO BUILDING IN SHIPPING SPLITS. 5. PROVIDE ECM IMPELLER FANS MOTORIZED IMPELLER CONTROL PANELS WITH H-O-A. 6. PROVIDE A SINGLE POINT OF CONNECTION FOR FANS AND WHEEL. 7. PROVIDE SERVICE LIGHTS IN THE FAN AND WHEEL SECTIONS WITH EXTERIOR MOUNTED SWITCHES.										8. PROVIDE SINGLE POINT OF CONNECTION FOR LIGHTS. 9. PROVIDE DOUBLE WALL CONSTRUCTION. 10. PROVIDE COASTAL CORROSION PROTECTION ON THE COILS. 11. PROVIDE MINIMUM SCCR OF 10,000 AIC. 12. PROVIDE WITH REFRIGERANT LEAK DETECTION SYSTEM.												

SPLIT SYSTEM AIR COOLED CONDENSING (OUTDOOR) UNIT SCHEDULE

UNIT NO	SERVES	COOLING LOAD MBH	REFRIG TYPE	SEER	DESIGN TEMP °F DB/WB	CONDENSER	COMPRESSOR(S)	RS/RL LINESET TUBING SIZES (OUSIDE Ø)	ELECTRICAL			BASIS OF DESIGN	NOTES
						FAN(S) QTY	QTY		VOLTS/ PHASE	MCA	MOCP		
OU-1	IU-1	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-2	IU-2	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-3	IU-3	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUY-AK12NL	1,2,4,5,6,7,8,10
OU-4	IU-4	12	R-454B	21.1	90.0/72.0	1	1	1/2" / 1/4"	208/1	16	27	TRANE-MITSUBISHI PUZ-AK12NL	1,3,4,5,6,7,8,10
CU-1	BCU-1	48	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	9.4	15	TRANE 5TWA4048A4	3,4,5,6,7,8,9,10
CU-2	BCU-2	42	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	8.6	15	TRANE 5TWA4042A4	3,4,5,6,7,8,9,10
CU-3	BCU-3	48	R-454B	14.0	90.0/72.0	1	1	7/8" / 5/16"	460/3	9.4	15	TRANE 5TWA4048A4	3,4,5,6,7,8,9,10
NOTES: 1. PROVIDE MANUFACTURER'S RECOMMENDED 24" HIGH STAND. 2. COOLING ONLY - PROVIDE LOW AMBIENT PACKAGE TO COOL DOWN TO -20°F AMBIENT. 3. PROVIDE LOW AMBIENT PACKAGE TO COOL DOWN TO 30°F AMBIENT. 4. SPLIT SYS. MUST HAVE AT LEAST A MAX. 150 FT. REFRIGERANT PIPING LENGTH & 50 FEET PIPING HEIGHT DIFFERENCE CAPACITY. 5. CORROSION RESISTANCE RATING OF 2000 HRS. OF DURABILITY UNDER ASTM 117 TESTING STANDARDS.													
6. PROVIDE MANUFACTURERS RECOMMENDED PRE-INSULATED REFRIGERANT LINESET SIZES 7. PROVIDE MANUFACTURED ENCLOSURE FOR EXPOSED INSULATED REFRIGERANT LINESET AND POWER CABLE FOR SPLIT SYSTEM 8. PROVIDE FIELD INSTALLED CRANK CASE HEATER, INSTALL IN ACCORDANCE WITH MANUFACTURER'S WRITTEN INSTRUCTIONS 9. HEAT PUMP 10. PROVIDE WITH EQUIPMENT MANUFACTURER'S VIBRATION ISOLATORS.													

EXHAUST FAN SCHEDULE

UNIT NO	SERVES	CFM	ESP IN WC	DRIVE TYPE	FAN TYPE	WEIGHT (LB)	FAN RPM	SONES	MAX BHP	HP	VOLTS/P HASE	BASIS OF DESIGN	ACCESSORIES
EF-1	MECH 115	650	0.35	DIRECT	SIDE WALL	56	1595	7.8	0.09	1/10	120/1	GREENHECK CUE - 090 - VG	A,B,C
TF-1	ELEC 114	350	0.25	DIRECT	INLINE	34	1234	5.0	0.03	1/6	120/1	GREENHECK SQ - 90 - VG	A,C
ACCESSORIES:		A. ELECTRONICALLY COMMUTATED MOTOR ECM. B. PROVIDE WITH LOW LEAKAGE, INSULATED MOTOR OPERATED DAMPER, WITH THERMALLY BROKEN INSULATED FRAME. C. MFR FAN MOUNTED DISCONNECT SWITCH.											

PUMP SCHEDULE

UNIT NO	SERVES	LOCATION	TYPE	GPM	TOTAL HEAD	PUMP EFF (%)	MOTOR DATA				SUCTION/ DISCHARGE (IN)	BASIS OF DESIGN	NOTES
							BHP	HP	RPM	VOLTS/PHASE			
P-1A	HW CIRCULATION	MECH 115	INLINE	133.6	69.5	65.7	3.5	7.5	1800	460/3	2.5/2.5	BELL & GOSSETT E-80	1
P-1B	HW CIRCULATION	MECH 115	INLINE	133.6	69.5	65.7	3.5	7.5	1800	460/3	2.5/2.5	BELL & GOSSETT E-80	1
NOTES: 1. PROVIDE WITH VARIABLE FREQUENCY DRIVE.													

STEAM-TO-WATER CONVERTER SCHEDULE

UNIT NO	SERVES	LOCATION	STEAM		HOT WATER				FOULING FACTOR	HEATING MBH	BASIS OF DESIGN	NOTES
			INLET PRESSURE	LB/HR	FLOW GPM	EWT °F	EWF °F	MAX WPD				
HX-1	BLDG 60 HYDRONIC HEAT SYST	MECH 115	5 PSI	2030	133.8	180	150	4.0 FT	0.0005	1963	BELL & GOSSETT SU-104-4	1
NOTES: 1. PROVIDE WITH OVERFLOW ALARM OUTPUTS.												

STEAM PRESSURE REDUCING VALVE SCHEDULE

TAG NO	LOCATION	CAPACITY LB/HR	INLET PRESSURE (PSIG)	OUTLET PRESSURE (PSIG)	VALVE SIZE	SAFETY VALVE		BASIS OF DESIGN	NOTES
						TAG	CAPACITY W/30 PSIG SET PRESS LB/HR		
PRV-1	MECH 115	2030	100	15	1"	SV-1	4631	LESLIE GPKP-1	1,2
NOTES: 1. CONTROL NOISE LEVEL TO BELOW 80 dBA BY PROVIDING ATTENUATION EQUIPMENT AS REQUIRED. 2. PRESSURE REDUCING VALVE MUST HAVE A 100 TO 1 PRESSURE TURN DOWN RATIO CAPACITY, OR PROVIDE PARALLEL 1/3 - 2/3 PRESSURE REDUCING VALVE ASSEMBLIES (2) AND SAFETY VALVES (2) FOR PRVS WITH SMALLER TURNDOWN CAPABILITY.									

AIR COOLED HEAT PUMP CONDENSING UNIT SCHEDULE FOR DOAS-1

UNIT NO	SERVES	NOMINAL COOLING CAPACITY FULL LOAD	LOW TEMP HEATING CAPACITY (MBH)	REFRIG TYPE	REFRIG Ø LINE SIZES (IN) RS/RL	EER FULL LOAD	DESIGN TEMP	CONDENSER FAN(S)				COMPRESSOR(S)			CAPACITY STEPS	ELECTRICAL				BASIS OF DESIGN	NOTES
							COND AMBIENT	QTY	FAN HP EA	FLA EA	LRA EA	QTY	FLA EA	LRA EA		VOLTS/PHASE	MCA	MDP	WEIGHT		
CU-4	DOAS-1	10.0	-	R-454B	1-3/8" / 1/2"	10.8	90° F	2	1.0	2.2	10.1	2	17.8	140	2	460/3	46	60	950LB	TRANE ODYSEY TWA 240K 4DAA	1,2,3,4,5,6,7,8
NOTES:	1.	DUAL COMPRESSORS/ DUAL CIRCUIT. THE LEAD COMPRESSOR MUST BE A DIGITAL SCROLL COMPRESSOR WITH A 25%-100% CAPACITY OPERATING RANGE. NOTE: BASIS OF DESIGN UNIT IS FACTORY MODIFIED TO ACHIEVE THIS REQUIREMENT.																			
	2.	COASTAL CORROSION LEVEL - COATED COMMERCIAL COIL PROTECTION																			
		3. MANUFACTURERS MICRO PROCESSED CONTROLS; BOD; TRANES4MB10 7000																			
		4. HAIL GUARDS, SPRING VIBRATION, ISOLATORS, BACNET INTERFACE																			
		5. REFRIGERANT LEAK DETECTION SYSTEM																			
		6. PROVIDE MINIMUM SCCR OF 10,000 A/C.																			
		7. HEAT PUMP																			
		8. BASIS OF DESIGN UNIT WITHIN THE SPACE CONSTRAINTS OF THE EQUIPMENT ARE SHOWN.																			

SPLIT SYSTEM INDOOR UNIT (IU) SCHEDULE

UNIT NO	SERVES	HEATING LOAD (MSH)	REFRIG TYPE	NOMINAL COOLING TONS	HEATING CAPACITY @4°F (MBH)	CABINET TYPE	NOMINAL CFM	SEER	ELECTRICAL VOLTS/PH	WEIGHT LBS	BASIS OF DESIGN	NOTES
IU-1	ELEV MACHINE RM	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-2	NMCI 110	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-3	NMCI 209B	N/A	R-454B	1.0	N/A	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	1,3,4
IU-4	OFFICE 108	12	R-454B	1.0	7.0	WALL MOUNTED CASSETTE	320	20.8	208/1	30	TRANE-MITSUBISHI PKA-AL12NL	2,3,4
NOTES:		1. COOLING ONLY UNIT. 2. HEAT PUMP UNIT. 3. PROVIDE WITH MANUFACTURER'S DELUXE WIRED, WALL MOUNTED CONTROLLER (BOD: TRANE-MITSUBISHI TAR-42MAAUS). 4. PROVIDE PRE-MANUFACTURED ENCLOSURE FOR EXPOSED PIPING WITHIN SERVED ROOM.										

STEAM CONDENSATE RECIEVER AND RETURN PUMP SCHEDULE

UNIT NO	LOCATION	CAPACITY		DISCHARGE PRESSURE (PSIG)	RECEIVER CAPACITY (GAL)	MOTOR DATA		RPM	BASIS OF DESIGN	NOTES
		LB/HR	GPM			HP	VOLTS/ PHASE			
CR-1	MECHANICAL 115	2100	9.0	75	23	3.0	460/3	3500	BELL & GOSSETT MODEL 23 CB9 - 75	1,2,3,4,5,6,7
NOTES: 1. PROVIDE DUPLEX PUMP SET. 2. PROVIDE FACTORY MOUNTED CONTROL PANEL, (NEMA 2 ENCLOSURE) FLOAT SWITCHES, AND ELECTRIC ALTERNATOR. 3. PROVIDE WATER LEVEL GAGE WITH SHUT OFF VALVE. 4. PROVIDE DISCHARGE PRESSURE GAUGE. 5. PROVIDE HIGH LEVEL ALARM. 6. PROVIDE DUPLEX DISCHARGE MANIFOLD. 7. SEE SPECIFICATIONS FOR FURTHER DETAILS.										

FINTUBE RADIATION SCHEDULE

UNIT NO	ELEMENT				ENCLOSURE		CAPACITY BTU/HLF	AVERAGE WATER TEMP °F	BASIS OF DESIGN	NOTES
	TUBE SIZE	ROWS	FINS PER FOOT	SIZE (IN)	MOUNTING HEIGHT (IN)	ENCLOSURE HEIGHT (IN)				
FR-1	3/4	1	40	3-5/8 x 4-1/4	18	14	1040	170	VULCAN LV4-S14	1
FR-2	3/4	1	50	3-5/8 x 4-1/4	79 1/2	19 1/2	970	170	VULCAN LV4-DS19	2
NOTES: 1. SLOPE TOP STYLE 2. DOUBLE SLOPE TOP STYLE										

LOUVER SCHEDULE

UNIT NO	SERVES	TYPE	CFM	DIMENSIONS (INCHES)			MIN FREE AREA SQUARE FEET	BASIS OF DESIGN	NOTES
				LENGTH	HEIGHT	DEPTH			
L-1	MECH 115 MAKE UP	FIXED	650	36	36	6	0.77	RUSKIN/EUF6375DX	1,2,3,4,5,6
NOTES:	1. PROVIDE WITH BIRDSCREEN. 2. LOUVER SHALL BE DRAINABLE. 3. PROVIDE WITH COASTAL CORROSION PROTECTION FINISH IN CUSTOM COLOR SELECTED BY CONTRACTING OFFICER.			4. PROVIDE LOUVER RATED FOR MINIMUM 35 PSF WIND LOAD. 5. PROVIDE WITH FLANGE OPTION. 6. LOUVER REPLACE LOWER WINDOW SASH. VERIFY IN FIELD SIZE OF LOUVER HEIGHT TO REPLACE LOWER WINDOW SASH.					